Mid Sweden University (Mittuniversitetet)

Omnisound

User-Centred Design and Evaluation
of a Unified Audio Interface

DT168G / IU127G
Human-Computer Interaction / Interaction Design

Project group: Group 1

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Abstract

TODO: Write an abstract of approximately 250 words after the results and discussion are complete. It should summarise the project's background, purpose, method, results and conclusions.

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Chapter 1

Introduction

1.1 Background

Music, podcasts and audiobooks can form part of the same person's listening activities, while being accessed through different applications. In such a situation, finding audio involves choosing both what to listen to and where to look for it. A listener may save an item in one application, continue a different item elsewhere and later need to remember which application holds each one. These possible organisational difficulties motivate the project.

Finding content is only one part of this problem. Saving an interesting item should also make it possible to find that item again. Similarly, returning to an unfinished podcast or audiobook requires a recognisable route back to the content and its listening position. For example, a student who stops listening to an audiobook during a study break should be able to identify the unfinished book and continue it later.

Combining the three audio types could provide a common place for these activities, but it also raises a design question. A mixed collection may be difficult to navigate if its content types are unclear or if too many items compete for attention. Music tracks, podcast episodes and audiobook chapters also need to remain identifiable within a shared interface. Common navigation and saving controls therefore have to be balanced with clear categories and information about each item.

Existing services already take different approaches to this organisation. Spotify combines music, podcasts and audiobooks in one service, while keeping the content types identifiable within its listening experience [1]. Apple offers separate applications for music, podcasts and audiobooks through Apple Music, Apple Podcasts and Apple Books [2]. These examples show that combining audio content is not a new idea, and neither approach is automatically easier for every user. Omnisound instead explores a specific interface question: whether shared search, a common library and clear content categories can make it easier to find saved material and continue unfinished listening without confusing the different audio types.

Omnisound is the project's response to this question: a web-interface prototype for finding, saving and resuming audio in one organised place. It primarily targets students

who regularly listen to music, podcasts or audiobooks, including listeners who use several content types and those who mainly listen to music. The project investigates how the interface can help these listeners keep track of saved and unfinished content.

The initial high-fidelity prototype (V1) is designed primarily for a desktop browser. It provides a shared search, content-type filters, a library of saved items and a Continue Listening area. Its two main task flows are finding and saving an item, then locating it in the library, and resuming an unfinished podcast or audiobook. These flows give the project a specific interaction focus within the broader audio concept.

1.2 Purpose

The purpose of the project is to investigate how a unified audio interface can support finding, organising and resuming different types of audio content while keeping the experience clear and manageable for the intended users. The motivation is to explore whether a common search and library can reduce the effort of returning to wanted content without making the differences between music, podcasts and audiobooks harder to understand.

This investigation also provides an opportunity to apply user-centred design within the HCI course. The course project requires an iterative process and usability testing in at least one iteration [3, pp. 2–3]. For Omnisound, the purpose is therefore pursued through requirements, alternative designs, prototypes and evaluation. The desired qualities are ease of learning, clear organisation and useful feedback; whether the design supports these qualities must be assessed through the project work.

1.3 Goals

The completed project should provide an identified set of user needs and a prioritised set of interface requirements. The requirements should make clear which listening activities the interface is intended to support and how they relate to the information collected from the target group. This provides a basis for assessing the design beyond its visual appearance.

The design deliverables should include documented alternatives, low-fidelity sketches and interactive high-fidelity prototypes. Alternatives should cover the organisation of audio categories and the presentation of content, such as category tabs compared with a mixed view with filters, and cards compared with lists. The sketches should preserve the early ideas, while the V1 and V2 prototypes should make the selected task flows available for examination. Together, these materials should show how the interface developed and why particular solutions were chosen.

A further deliverable is a documented usability evaluation of the important tasks: finding and saving content, locating saved content and returning to unfinished spoken audio. The evaluation should provide observations of task completion and difficulties, together with an account of how those observations informed design decisions. A record

of design iterations and reasoned conclusions should connect the requirements, prototypes and evaluation. These goals concern identifiable project outputs; they do not assume that all proposed changes will prove effective.

1.4 Delimitations

Omnisound is an interface prototype. The project focuses on navigation, content organisation and the visibility of actions and listening progress. It does not implement music-streaming infrastructure, integration with commercial audio services, payment or subscription processing, licensing systems, backend databases or production account management. The prototype is implemented only as a front end. Limiting the implementation in this way allows the design work to concentrate on the selected interactions within the course project.

Search uses a sample catalogue, and the player controls and displayed progress represent the intended listening experience. They are sufficient to explore how users move through the interface, recognise content and interpret feedback. They do not demonstrate reliable audio delivery, synchronisation between devices or the operation of a real streaming service. Conclusions must therefore concern interaction with the prototype rather than the technical performance of a complete product.

The task scope is also limited. Search and save, retrieval from the library and continuation of unfinished podcasts or audiobooks receive priority. Other possible activities, such as managing a production account or exploring a large commercial catalogue, are outside this evaluation focus. The primary context is desktop use, so the project cannot establish usability across all mobile devices or listening environments. Likewise, conclusions drawn from evaluation with the intended student group should be considered in relation to that group and the tasks tested.

Following the first usability test, a second version of Omnisound was developed to address the problems identified during evaluation. The changes focused on reducing the size of the Home banner, making Continue Listening easier to discover, and improving the clarity of the Save/Saved control. A second usability test was then conducted using V2 and the same three core tasks, allowing the group to compare the two iterations. The testing procedures and findings are presented in the Method and Results chapters. Since Omnisound remains an interface prototype, the conclusions concern the tested interactions rather than the technical performance of a complete streaming service.

1.5 Division of Work

The project was carried out by Obid Alrhman Aljabi, Yacoub Dawli and Sarshar Chavoshi Rad.

Yacoub Dawli was responsible for leading the interviews and moderating the usability tests. Obid Alrhman Aljabi and Sarshar Chavoshi Rad were responsible for documenting

participant responses and observations. During usability testing, Obid focused on participant behaviour and comments, while Sarshar documented task completion times and navigation steps.

The group collaborated on analysing user needs, discussing requirements and evaluating different interface designs. The design process included low-fidelity sketches, two high-fidelity prototypes and usability evaluations. Findings from the first evaluation informed the changes introduced in the second prototype.

The project findings and design decisions were discussed within the group and formed the basis for the report.

Chapter 2

Theory

2.1 User-Centred Design and Iteration

User-centred design begins with understanding the people who will use an interface, what they want to achieve and the circumstances in which they will use it. Rogers, Sharp and Preece explain that understanding people's activities and differences helps designers question assumptions about their needs [4, pp. 16–17]. A target group therefore provides a starting point for research, rather than a reason to assume that all its members have the same habits or expectations.

User needs and goals guide the requirements for an interface. Designers can then explore alternative ways to support those goals and represent the ideas in prototypes. Evaluation with intended users examines whether a proposed solution works as expected. A prototype is a candidate solution that can be tested by observing people using it [5, Sec. "What Is a Testable Prototype?"]. This makes evaluation part of design, rather than a check reserved for the finished product.

Iteration connects evaluation to further design decisions. Findings can lead to revised requirements, a different alternative or changes to an existing interface, followed by another evaluation [5, Sec. "Why Test a Prototype?"]. Simply producing several versions does not establish this connection. A meaningful iteration should explain which problem motivated a change and examine whether the problem remains. This principle is relevant to Omnisound's student audience and its two evaluated high-fidelity prototype iterations.

2.2 Usability and User Experience

ISO 9241-11:2018 relates usability to specified users, goals and contexts of use through effectiveness, efficiency and satisfaction. These definitions are reproduced in ISO/TR 25060:2023 [6, Sec. 3.1]. Usability concerns what people can achieve with a system in a particular situation; it cannot be established from appearance alone.

Effectiveness concerns accurate and complete achievement of a goal, while efficiency concerns the resources used in relation to that outcome [6, Secs. 3.1.7–3.1.8]. In an audio

interface, effectiveness could involve successfully saving an item and finding it again. Efficiency concerns the time and effort required to reach that outcome. Successful completion and a straightforward route are thus different aspects of the same task. Satisfaction concerns how well the experience of use meets the user's needs and expectations [6, Sec. 3.1.9].

Learnability and utility are additional useful HCI concepts. Learnability concerns how easily a person can learn to use an interface, including getting started with unfamiliar controls. Utility concerns whether the available functions support what users need to do [4, p. 21]. For example, a clearly labelled search may be easy to learn, while a way to return to unfinished audio addresses a functional need. These concepts complement the three ISO dimensions; they are not additional components of the ISO definition of usability.

User experience includes people's perceptions and responses before, during and after interaction, including anticipated use [6, Sec. 3.1.15]. Satisfaction and frustration are relevant examples. Task completion alone cannot explain how an interaction feels. Together, these concepts provide a basis for examining searching, saving, retrieving and resuming audio without treating one measure as a complete assessment of the interface.

2.3 Interface Design Principles

Visibility means making the options needed for a task apparent to the user [4, p. 28]. An available function has limited value if people cannot discover it. In an audio interface, recognisable navigation and information about listening progress can help users identify where to go and what can be continued.

Feedback communicates the outcome of an action and the current system state [7, Sec. 1]. After saving an item, a visible confirmation can help the user understand that the action succeeded. Feedback should be timely and clear enough to support the next decision, such as continuing to browse or opening the saved collection.

Consistency means using similar terms, controls and behaviour for similar interactions, both within a product and in relation to familiar conventions [7, Sec. 4]. A shared saving action across different audio types can support predictable interaction. Consistency still allows relevant differences in content, such as episode titles and audiobook chapters.

Affordances concern the actions possible in the relationship between a person and an object. Signifiers are perceivable cues that communicate what can be done [8]. In a graphical interface, a button label, icon or visual treatment can signify an available action. A control's appearance and the action it actually supports should agree. This distinction helps explain why making an action available and making it understandable are separate design concerns.

2.4 Cognition and Mental Models

A mental model is a user's understanding of how a system works. It supports predictions about what an action will do and develops through previous experience [9, Sec. "What Are Mental Models?"]. Users and designers may have different models of the same interface. Familiarity with another audio service may, for example, shape expectations about where saved items belong or whether opening unfinished content will resume it. Clear category names and navigation can help communicate how a shared audio collection is organised.

Attention concerns which information people notice while pursuing a goal. Layout and visual prominence matter because competing information can reduce the visibility of relevant content [7, Sec. 8]. A large introductory area and a route back to unfinished listening may compete for attention on the same screen. This provides a reason to examine the relative prominence of elements, without assuming that size alone determines what every user notices.

Recognition over recall means displaying options and information that users can identify, rather than requiring them to remember details [7, Sec. 6]. Visible content categories, item titles and progress information can support this recognition. For an audio collection, these cues may help someone distinguish saved material from unfinished listening and choose a suitable next action.

2.5 User Research and Evaluation Methods

Qualitative interviews explore people's experiences, needs and expectations. Open questions and follow-up questions allow participants to explain what matters to them in their own terms [10]. Interviews can inform requirements, but they collect accounts of behaviour rather than direct observations of interface use. Combining these accounts with observations can provide a fuller basis for understanding users.

Affinity diagramming organises related observations or ideas into groups that express common themes [11, Sec. "Defining Affinity Diagramming"]. It can help a design team make sense of interview notes and identify connections between needs. The grouping is an interpretation of the material, so observations that differ from a common theme should also remain visible in the analysis.

Personas are research-informed representations of users' goals, needs and expectations. Qualitative personas draw on patterns in research to help a team focus its design decisions [12]. They summarise relevant differences between users rather than reproducing every interview as a separate profile. A persona based only on assumptions should not be treated as evidence about the intended audience, and a qualitative persona does not establish how common a user type is in the wider population.

Low-fidelity prototypes, such as paper sketches, represent early ideas with limited detail. They allow designers to explore layouts and navigation before investing in a detailed interface, and can be changed or discarded readily [5, Sec. "Benefits of Low-Fidelity Prototypes"]. More detailed, interactive prototypes allow evaluation of specific

controls and task flows [5, Sec. “Benefits of High-Fidelity Prototypes”]. The appropriate level of detail depends on the question being investigated; visual polish alone does not establish that a design meets users’ needs.

Think-aloud usability testing asks participants to express their thoughts while completing representative tasks with an interface [13, Sec. “Defining Thinking Aloud Testing”]. Observations of actions, hesitation and comments can reveal difficulties and help explain why a participant chose a particular route. These observations can inform redesign, including changes to the visibility or interpretation of controls.

Qualitative usability tests also have limitations. Speaking aloud creates an unusual situation, and prompts can influence behaviour [13, Sec. “Downsides of Think-Aloud”]. Findings from a small group can identify problems in the tested interactions, but do not automatically provide statistically generalisable conclusions. Task times and navigation steps should therefore be interpreted alongside observations and the test context, rather than treated as universal measures of performance.

2.6 Research Ethics

User research should respect participants’ ability to make an informed and voluntary decision. Vetenskapsrådet’s *God forskningssed 2024* explains that people should understand what participation involves, how information will be handled and how to withdraw consent [14, pp. 62–63]. Information should be understandable to the intended participants, who should have an opportunity to ask questions and decline or stop participation without pressure.

Privacy requires responsible handling of participant information. Only information needed for the study should be collected, and it should be protected appropriately. Removing names alone may not prevent identification when other details can be combined [14, pp. 66–67]. This matters when selecting background details, quotations and observations for a report.

Accurate reporting is also an ethical responsibility. Observations and measurements should be presented honestly, with interpretations and limitations made clear [14, pp. 11–12]. Results that challenge a design should be retained rather than adjusted to support a preferred conclusion. These principles provide the basis for describing the group’s documented ethical procedures in the Method chapter.

Chapter 3

Method

TODO: Chapter not yet written.

Chapter 4

Design

TODO: Chapter not yet written.

Chapter 5

Results

TODO: Chapter not yet written.

Chapter 6

Discussion

TODO: Chapter not yet written.

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